

Lecture 2 - Discrete Systems

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- Difference Equations (DEQs) are used to model discrete systems.
 - DEQs allow us to express systems as the linear combinations of other systems.
- The goal of discrete system design is to select optimal values of the α_k and β_k terms to best model a given scenario.
- Systems without feedback (no $y(n-k)$, $k \geq 1$ terms) exhibit both *linearity* and *time-invariance*.
 - A system exhibits linearity if its output, for some given input, can be expressed as the linear combination of the outputs of other systems with the same input.
 - A system exhibits time-invariance if delaying the input to the system results in the same output but delayed by the same amount.

1 Find the outputs of the system $y(n) = x(n) + \frac{1}{2}x(n-1) + \frac{1}{4}x(n-2)$ for the following inputs:

Note that $y(n)$ has no feedback and thus exhibits both linearity and time-invariance

1.a $x_a(n) = \begin{cases} 1, & \text{if } n = 0 \\ 0, & \text{otherwise} \end{cases}$

n	0	1	2	3	4	5
$x_a(n)$	1	0	0	0	0	0
$y(n)$	1	$\frac{1}{2}$	$\frac{1}{4}$	0	0	0

1.b $x_b(n) = \begin{cases} 1, & \text{if } n = 2 \\ 0, & \text{otherwise} \end{cases}$

n	0	1	2	3	4	5
$x_b(n)$	0	0	1	0	0	0
$y(n)$	0	0	1	$\frac{1}{2}$	$\frac{1}{3}$	0

Notice here that $x_b(n) = x_a(n-2)$, and thus $x_b(n)$ is the same as $x_a(n)$ but delayed by 2 cycles. The fact that our output in (b) is likewise delayed by 2 cycles as compared to in (a) is corroboration of the fact that $y(n)$ exhibits time-invariance.

$$1.c \quad x_c(n) = \begin{cases} 2, & \text{if } n = 1 \\ 0, & \text{otherwise} \end{cases}$$

n	0	1	2	3	4	5
$x_c(n)$	0	2	0	0	0	0
$y(n)$	0	2	1	$\frac{1}{2}$	0	0

Notice here once again that $x_c(n) = 2x_a(n-1)$ and thus $x_c(n)$ is the same as the input in $x_a(n)$ but multiplied by 2 and delayed by a cycle. The fact that $y(n)$ is now delayed by 1 cycle and twice that as in (a) speaks to the fact $y(n)$ exhibits both linearity and time-invariance.

$$1.d \quad x_d(n) = \begin{cases} 1, & \text{if } n = 0, 2 \\ 2, & \text{if } n = 1 \\ 0, & \text{otherwise} \end{cases}$$

n	0	1	2	3	4	5
$x_d(n)$	1	2	1	0	0	0
$y(n)$	1	$\frac{5}{2}$	$\frac{9}{4}$	$\frac{3}{2}$	$\frac{3}{4}$	0

Notice again that $x_d(n) = x_a(n) + x_b(n) + x_c(n)$, thus $y(n)$ is just the sum of all of the $y(n)$'s computed for (a), (b), and (c).

- The state-equations are an alternative representation to DEQs to represent systems.
 - Also called the (A, B, C, D) representation.
 - In its general form, the state-space representation is written as two equations:
 - * $\overrightarrow{S(n+1)} = \mathbf{A}\overrightarrow{S(n)} + \mathbf{B}(n)x(n)$
 - * $y(n) = \mathbf{C}\overrightarrow{S(n)} + \mathbf{D}x(n)$
 - $\overrightarrow{S(n)}$ represents the system's current state.
 - $\overrightarrow{S(n+1)}$ represents the system's next state.
 - $\overrightarrow{S(0)}$ is always defined to be zero.
 - $\overrightarrow{S(0)}$ is called the system's *zero state*.
 - Systems in state-space form have their outputs described as a function of the system's current state $\overrightarrow{S(n)}$ and current input $x(n)$.
 - $\overrightarrow{S(n)}$ is found by renaming the inputs/feedbacks in the difference equation representation of $y(n)$ such that the entire difference equation can be expressed in terms of $x(n)$ and $S(n)$.

2 Express the system $y(n) = x(n) + \frac{1}{2}x(n-1) + \frac{1}{4}x(n-2)$ in state-space form

2.I Find $\overrightarrow{S(n)}$

Let:

$$s_1(n) = x(n-1)$$

$$s_2(n) = x(n-2)$$

Then:

$$s_1(n+1) = x(n)$$

$$s_2(n+1) = x(n-1) = s_1(n)$$

Thus, let $\overrightarrow{S(n)} = \begin{pmatrix} s_1(n) \\ s_2(n) \end{pmatrix}$.

2.II Find $\overrightarrow{S(n+1)}$

$$\begin{aligned}\overrightarrow{S(n+1)} &= \begin{pmatrix} s_1(n+1) \\ s_2(n+1) \end{pmatrix} \\ &= \begin{pmatrix} x(n) \\ s_1(n) \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(n)\end{aligned}$$

Therefore, our A and B matrices are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ respectively.

2.III Find $y(n)$

$$\begin{aligned}y(n) &= x(n) + \frac{1}{2}x(n-1) + \frac{1}{4}x(n-2) \\ &= \frac{1}{2}s_1(n) + \frac{1}{4}s_2(n) + x(n) \\ &= \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)\end{aligned}$$

Therefore, our C and D matrix are $\begin{pmatrix} \frac{1}{2} & \frac{1}{4} \end{pmatrix}$ and $\begin{pmatrix} 1 \end{pmatrix}$ respectively.

Putting it all together, our system can be represented in state space form as:

$$\begin{aligned}\overrightarrow{S(n+1)} &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} x(n) \\ y(n) &= \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)\end{aligned}$$

3 Represent the system $y(n) = x(n) - x(n-1) + y(n-2)$ in state-space form

3.I Find $\overrightarrow{S(n)}$

Let:

$$s_1(n) = x(n-1)$$

$$s_2(n) = y(n-2)$$

$$s_3(n) = y(n-1)$$

Then:

$$s_1(n+1) = x(n)$$

$$s_2(n+1) = y(n-1) = s_3(n)$$

$$s_3(n+1) = y(n) = x(n) - x(n-1) + y(n-2) = -s_1(n) + s_2(n) + x(n)$$

Thus, let $\overrightarrow{S(n)} = \begin{pmatrix} s_1(n) \\ s_2(n) \\ s_3(n) \end{pmatrix}$.

3.II Find $\overrightarrow{S(n+1)}$

$$\begin{aligned}
 \overrightarrow{S(n+1)} &= \begin{pmatrix} s_1(n+1) \\ s_2(n+1) \\ s_3(n+1) \end{pmatrix} \\
 &= \begin{pmatrix} x(n) \\ s_3(n) \\ -s_1(n) + s_2(n) + x(n) \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} x(n)
 \end{aligned}$$

Therefore, our A and B matrices are $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ respectively.

3.III Find $y(n)$

$$\begin{aligned}
 y(n) &= x(n) - x(n-1) + y(n-2) \\
 &= -s_1(n) + s_2(n) + x(n) \\
 &= \begin{pmatrix} -1 & 1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)
 \end{aligned}$$

Therefore, our C and D matrices are $\begin{pmatrix} -1 & 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \end{pmatrix}$ respectively.

Putting it all together, our system can be represented in state space form as:

$$\begin{aligned}
 \overrightarrow{S(n+1)} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} x(n) \\
 y(n) &= \begin{pmatrix} -1 & 1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)
 \end{aligned}$$